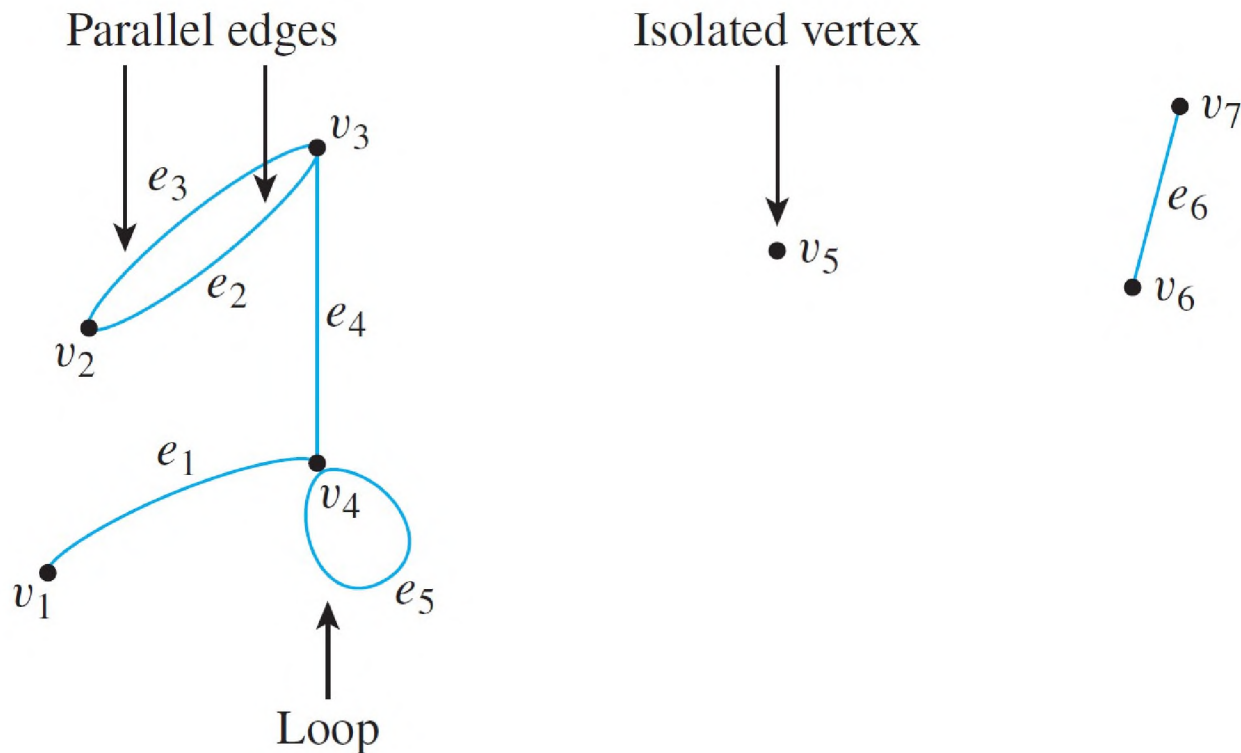


GRAPHS AND TREES



Graphs: Definitions and Basic Properties

The edges may be straight or curved and should either connect one vertex to another or a vertex to itself, as shown below.





Graphs: Definitions and Basic Properties

In this drawing, the vertices have been labeled with v 's and the edges with e 's.

When an edge connects a vertex to itself (as e_5 does), it is called a *loop*. When two edges connect the same pair of vertices (as e_2 and e_3 do), they are said to be *parallel*.

It is quite possible for a vertex to be unconnected by an edge to any other vertex in the graph (as v_5 is), and in that case the vertex is said to be *isolated*.



Graphs: Definitions and Basic Properties

The formal definition of a graph follows.

- **Definition**

A **graph** G consists of two finite sets: a nonempty set $V(G)$ of **vertices** and a set $E(G)$ of **edges**, where each edge is associated with a set consisting of either one or two vertices called its **endpoints**. The correspondence from edges to endpoints is called the **edge-endpoint function**.

An edge with just one endpoint is called a **loop**, and two or more distinct edges with the same set of endpoints are said to be **parallel**. An edge is said to **connect** its endpoints; two vertices that are connected by an edge are called **adjacent**; and a vertex that is an endpoint of a loop is said to be **adjacent to itself**.

An edge is said to be **incident on** each of its endpoints, and two edges incident on the same endpoint are called **adjacent**. A vertex on which no edges are incident is called **isolated**.



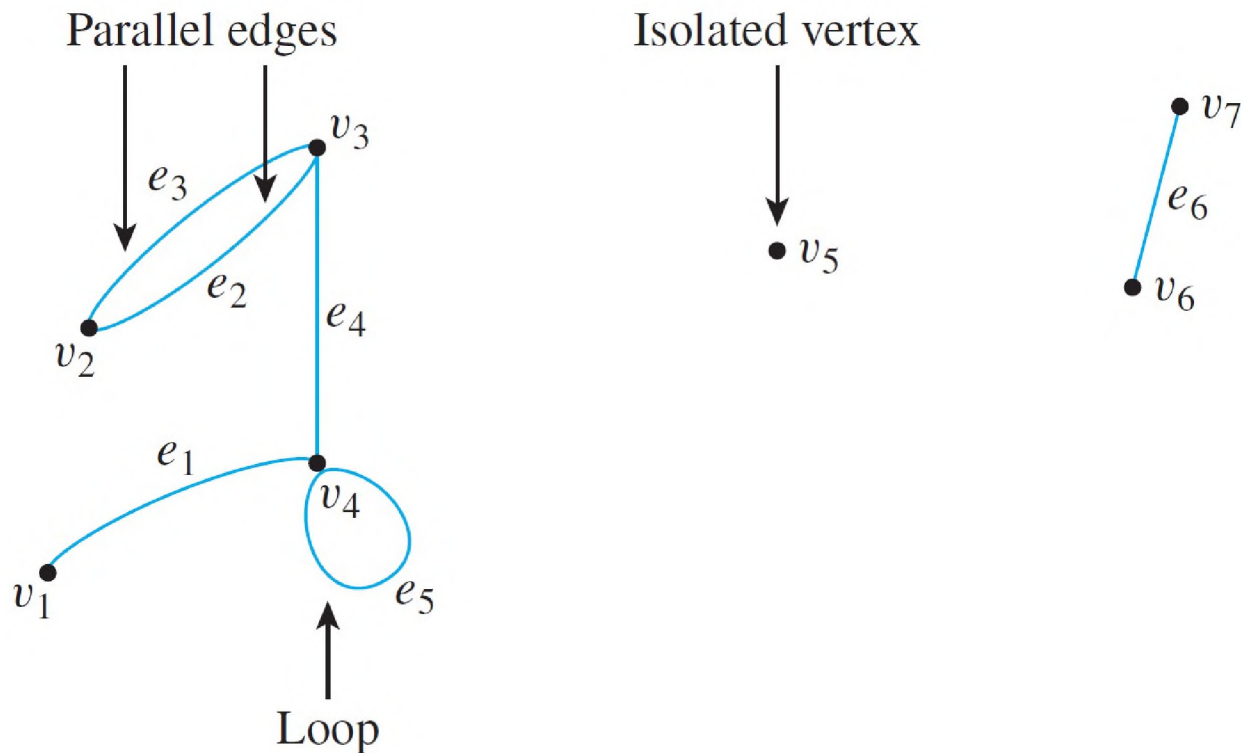
Graphs: Definitions and Basic Properties

Graphs have pictorial representations in which the vertices are represented by dots and the edges by line segments.

A given pictorial representation uniquely determines a graph.

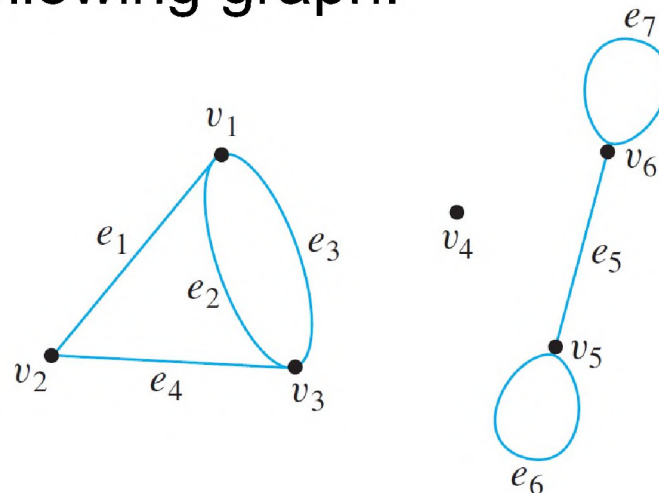
Graphs: Definitions and Basic Properties

The edges may be straight or curved and should either connect one vertex to another or a vertex to itself, as shown below.



Example 1 – Terminology

Consider the following graph:



- Write the vertex set and the edge set, and give a table showing the edge-endpoint function.
- Find all edges that are incident on v_1 , all vertices that are adjacent to v_1 , all edges that are adjacent to e_1 , all loops, all parallel edges, all vertices that are adjacent to themselves, and all isolated vertices.



Example 1(a) – *Solution*

vertex set = $\{v_1, v_2, v_3, v_4, v_5, v_6\}$

edge set = $\{e_1, e_2, e_3, e_4, e_5, e_6, e_7\}$

edge-endpoint function:

Edge	Endpoints
e_1	$\{v_1, v_2\}$
e_2	$\{v_1, v_3\}$
e_3	$\{v_1, v_3\}$
e_4	$\{v_2, v_3\}$
e_5	$\{v_5, v_6\}$
e_6	$\{v_5\}$
e_7	$\{v_6\}$



Example 1(a) – *Solution*

cont'd

Note that the isolated vertex v_4 does not appear in this table.

Although each edge must have either one or two endpoints, a vertex need not be an endpoint of an edge.



Example 1(b) – *Solution*

cont'd

e_1 , e_2 , and e_3 are incident on v_1 .

v_2 and v_3 are adjacent to v_1 .

e_2 , e_3 , and e_4 are adjacent to e_1 .

e_6 and e_7 are loops.

e_2 and e_3 are parallel.

v_5 and v_6 are adjacent to themselves.

v_4 is an isolated vertex.



Graphs: Definitions and Basic Properties

- **Definition**

A **directed graph**, or **digraph**, consists of two finite sets: a nonempty set $V(G)$ of vertices and a set $D(G)$ of directed edges, where each is associated with an ordered pair of vertices called its **endpoints**. If edge e is associated with the pair (v, w) of vertices, then e is said to be the **(directed) edge** from v to w .

Note that each directed graph has an associated ordinary (undirected) graph, which is obtained by ignoring the directions of the edges.



Special Graphs



Special Graphs

- Definition and Notation

A **simple graph** is a graph that does not have any loops or parallel edges. In a simple graph, an edge with endpoints v and w is denoted $\{v, w\}$.



Example 8 – *A Simple Graph*

Draw all simple graphs with the four vertices $\{u, v, w, x\}$ and two edges, one of which is $\{u, v\}$.

Solution:

Each possible edge of a simple graph corresponds to a subset of two vertices.

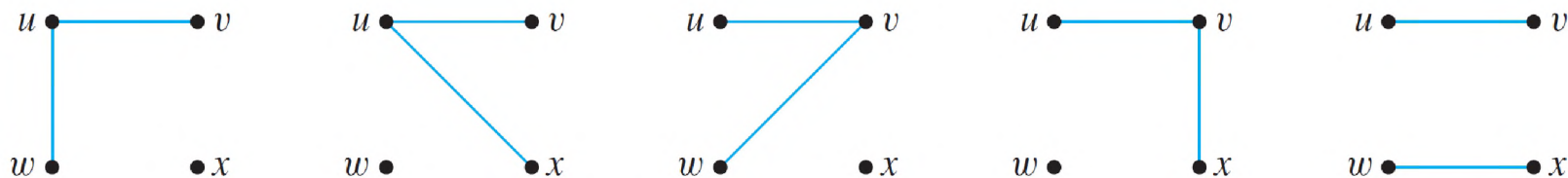
Given four vertices, there are $\binom{4}{2} = 6$ such subsets in all: $\{u, v\}$, $\{u, w\}$, $\{u, x\}$, $\{v, w\}$, $\{v, x\}$, and $\{w, x\}$.

Example 8 – *Solution*

cont'd

Now one edge of the graph is specified to be $\{u, v\}$, so any of the remaining five from this list can be chosen to be the second edge.

The possibilities are shown below.





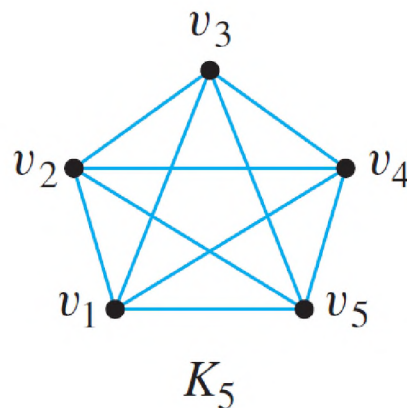
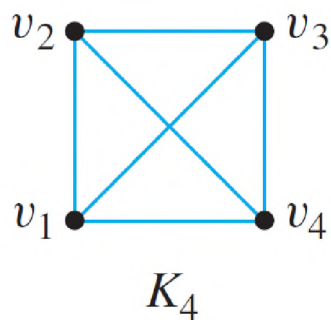
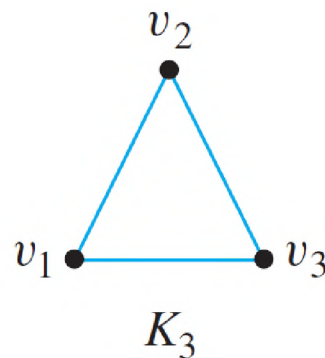
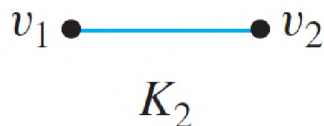
Special Graphs

- **Definition**

Let n be a positive integer. A **complete graph on n vertices**, denoted K_n , is a simple graph with n vertices and exactly one edge connecting each pair of distinct vertices.

Example 9 – Complete Graphs on n Vertices: K_1, K_2, K_3, K_4, K_5

The complete graphs K_1, K_2, K_3, K_4 , and K_5 can be drawn as follows:



Special Graphs

In yet another class of graphs, the vertex set can be separated into two subsets: Each vertex in one of the subsets is connected by exactly one edge to each vertex in the other subset, but not to any vertices in its own subset. Such a graph is called *complete bipartite*.

• Definition

Let m and n be positive integers. A **complete bipartite graph on (m, n) vertices**, denoted $K_{m,n}$, is a simple graph with distinct vertices $v_1, v_2, \dots, v_m$ and $w_1, w_2, \dots, w_n$ that satisfies the following properties: For all $i, k = 1, 2, \dots, m$ and for all $j, l = 1, 2, \dots, n$,

1. There is an edge from each vertex v_i to each vertex w_j .
2. There is no edge from any vertex v_i to any other vertex v_k .
3. There is no edge from any vertex w_j to any other vertex w_l .



Special Graphs

- **Definition**

A graph H is said to be a **subgraph** of a graph G if, and only if, every vertex in H is also a vertex in G , every edge in H is also an edge in G , and every edge in H has the same endpoints as it has in G .



The Concept of Degree



The Concept of Degree

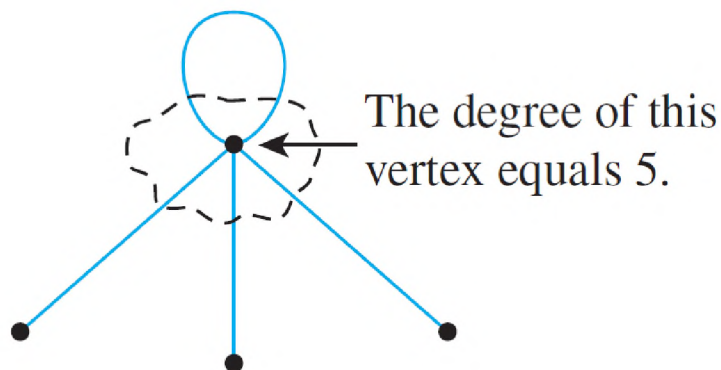
- **Definition**

Let G be a graph and v a vertex of G . The **degree of v** , denoted $\deg(v)$, equals the number of edges that are incident on v , with an edge that is a loop counted twice. The **total degree of G** is the sum of the degrees of all the vertices of G .

The Concept of Degree

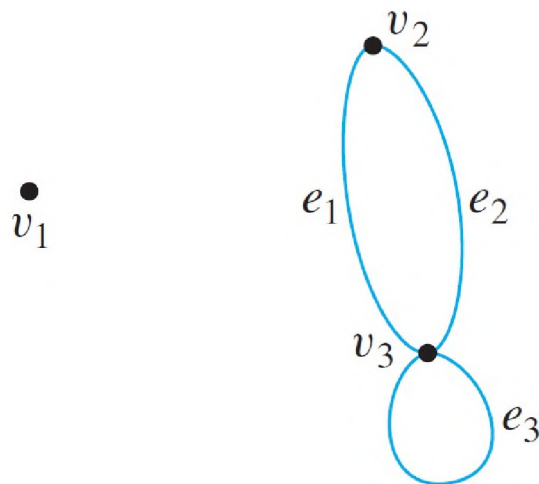
Since an edge that is a loop is counted twice, the degree of a vertex can be obtained from the drawing of a graph by counting how many end segments of edges are incident on the vertex.

This is illustrated below.



Example 12 – *Degree of a Vertex and Total Degree of a Graph*

Find the degree of each vertex of the graph G shown below. Then find the total degree of G .





Example 12 – *Solution*

$\deg(v_1) = 0$ since no edge is incident on v_1 (v_1 is isolated).

$\deg(v_2) = 2$ since both e_1 and e_2 are incident on v_2 .

$\deg(v_3) = 4$ since e_1 and e_2 are incident on v_3 and the loop e_3 is also incident on v_3 (and contributes 2 to the degree of v_3).

total degree of $G = \deg(v_1) + \deg(v_2) + \deg(v_3)$

$$= 0 + 2 + 4$$

$$= 6.$$



The Concept of Degree

Note that the total degree of the graph G of Example 12, which is 6, equals twice the number of edges of G , which is 3.



The Concept of Degree

Theorem 10.1.1 The Handshake Theorem

If G is any graph, then the sum of the degrees of all the vertices of G equals twice the number of edges of G . Specifically, if the vertices of G are $v_1, v_2, \dots, v_n$, where n is a nonnegative integer, then

$$\begin{aligned}\text{the total degree of } G &= \deg(v_1) + \deg(v_2) + \dots + \deg(v_n) \\ &= 2 \cdot (\text{the number of edges of } G).\end{aligned}$$

Corollary 10.1.2

The total degree of a graph is even.



The Concept of Degree

The following proposition is easily deduced from Corollary 10.1.2 using properties of even and odd integers.

Proposition 10.1.3

In any graph there are an even number of vertices of odd degree.